Olist Commerce360

An End-to-End E-Commerce Data Warehouse and Business Intelligence Platform

P1_04 - Fact and Dimension Identification

Member 1 - Warehouse & integration lead	[Name] - [Student ID]
Member 2 - Business analysis & modeling lead	[Name - proposed: Khadija] - [Student ID]
Member 3 - Analytics & BI lead	[Name] - [Student ID]
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Instructor	[Instructor name]
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1. Method

The model follows the four-step dimensional design process: (1) select the business process, (2) declare the grain, (3) identify the dimensions, (4) identify the facts (measures). Candidate tables were checked against the profiled source data and the business questions in P1_03. A table was kept only when it has a clear business purpose, a source, a unique grain and at least one question that needs it.

2. Business processes and fact tables

Business process	Fact table	Fact type	Why a separate fact
Sales (order line items)	fact_order_items	Transaction	Revenue, units and freight exist per item; product and seller are only known at item level.
Order fulfilment lifecycle	fact_orders	Accumulating snapshot	Five milestone dates per order; delivery measures and order counts must be counted once per order.
Payment transactions	fact_payments	Transaction	Payment type and instalments exist per payment record (up to 29 per order).
Customer reviews	fact_reviews	Transaction	Reviews have their own dates; 547 orders have more than one review and one review can cover several orders.

3. Candidate evaluation

Candidate	Decision	Reason
fact_order_items	Kept	Core sales fact; answers BQ-01, 02, 04, 13.
fact_orders	Kept	Order counts, status, delivery performance, repeat customers; BQ-01, 03, 07, 08, 09, 11, 12.
fact_payments	Kept	Payment mix and instalments; BQ-05, 06.
fact_reviews	Kept	Review-level satisfaction trend; BQ-10.
Single flattened sales table	Rejected	Mixes grains: a naive item x payment join creates 117,601 rows and overstates the amount paid by BRL 4.30 M.
dim_location / dim_geography	Rejected (folded)	Geography attributes are stored in dim_customer and dim_seller; a separate table would create a snowflake.
dim_category	Rejected (folded)	1:1 translation lookup; stored as attributes of dim_product.
dim_review	Rejected	Review text is free Portuguese text with almost one value per review; not a useful dimension. The review identifier is kept as a degenerate key.
Inventory snapshot / promotion facts	Rejected	No stock or promotion data in the source.

4. Fact tables

4.1 fact_order_items (Transaction fact)

Business process	Sales (order line items)
Grain	One row per product line within a customer order (order_id + order_item_id). Each row is one unit of one product sold by one seller.
Primary key	order_id + order_item_id (degenerate source identifiers)
Source	olist_order_items_dataset.csv joined to olist_orders_dataset.csv and olist_customers_dataset.csv for keys
Expected rows	112,650

Foreign keys

FK column	References	Meaning
order_purchase_date_key	dim_date.date_key	Date the order was placed.
shipping_limit_date_key	dim_date.date_key	Seller's deadline to hand the item to the carrier.
customer_key	dim_customer.customer_key	Customer who placed the order.
product_key	dim_product.product_key	Product sold.
seller_key	dim_seller.seller_key	Seller who sold the item.
order_status_key	dim_order_status.order_status_key	Current status of the parent order (used to exclude cancelled sales).

Measures

Measure	Type	Additivity	Definition
item_price	NUMERIC(10,2)	Additive	Selling price of the item (BRL), excluding freight.
freight_value	NUMERIC(10,2)	Additive	Freight charged for this item (BRL).
item_gross_value	NUMERIC(10,2)	Additive	item_price + freight_value.
item_quantity	SMALLINT	Additive	Units on this row (always 1 in source; repeated rows mean quantity > 1).

Degenerate dimensions: order_id, order_item_id.

4.2 fact_orders (Accumulating snapshot fact)

Business process	Order fulfilment lifecycle
Grain	One row per customer order (order_id), updated as the order passes purchase, approval, carrier hand-off and delivery milestones.
Primary key	order_id (degenerate source identifiers)
Source	olist_orders_dataset.csv + aggregates of order_items, order_payments and order_reviews per order
Expected rows	99,441

Foreign keys

FK column	References	Meaning
customer_key	dim_customer.customer_key	Customer who placed the order.
order_status_key	dim_order_status.order_status_key	Current order status.
order_purchase_date_key	dim_date.date_key	Order placed (role: purchase date).
approved_date_key	dim_date.date_key	Payment approved (role: approval date); -1 if not yet.
carrier_date_key	dim_date.date_key	Handed to carrier (role: carrier date); -1 if not yet.
delivered_date_key	dim_date.date_key	Delivered to customer (role: delivery date); -1 if not yet.
estimated_delivery_date_key	dim_date.date_key	Delivery date promised at purchase (role: estimated date).

Measures

Measure	Type	Additivity	Definition
item_count	SMALLINT	Additive	Number of item rows in the order (0 for orders without items).
order_item_value	NUMERIC(12,2)	Additive	Sum of item_price for the order.
order_freight_value	NUMERIC(12,2)	Additive	Sum of freight_value for the order.
order_payment_value	NUMERIC(12,2)	Additive	Total amount paid for the order (all payment records).
payment_count	SMALLINT	Additive	Number of payment records for the order.
days_purchase_to_approval	NUMERIC(7,2)	Non-additive	Elapsed days from purchase to payment approval.
days_approval_to_carrier	NUMERIC(7,2)	Non-additive	Elapsed days from approval to carrier hand-off (seller handling).
days_carrier_to_customer	NUMERIC(7,2)	Non-additive	Elapsed days in transit.
days_purchase_to_delivery	NUMERIC(7,2)	Non-additive	Total elapsed days from purchase to delivery.
days_promised	NUMERIC(7,2)	Non-additive	Promised lead time: purchase to estimated delivery.
delivery_delay_days	INTEGER	Non-additive	Delivered date minus estimated date in whole days (negative = early).
is_delivered	SMALLINT	Additive	1 if status = delivered and a delivery date exists, else 0.
is_late	SMALLINT	Additive	1 if delivered after the estimated date, 0 if on time; NULL if not delivered.
customer_order_number	SMALLINT	Non-additive	Sequence of this order among the customer's orders (1 = first purchase).
is_repeat_purchase	SMALLINT	Additive	1 if customer_order_number > 1.
review_count	SMALLINT	Additive	Number of reviews received for the order.
avg_review_score	NUMERIC(3,2)	Non-additive	Mean review score of the order (1-5); NULL if no review.

Degenerate dimensions: order_id, purchase_hour.

4.3 fact_payments (Transaction fact)

Business process	Payment transactions
Grain	One row per payment record of an order (order_id + payment_sequential). An order paid with a card and two vouchers has three rows.
Primary key	order_id + payment_sequential (degenerate source identifiers)
Source	olist_order_payments_dataset.csv joined to olist_orders_dataset.csv for customer, status and date
Expected rows	103,886

Foreign keys

FK column	References	Meaning
order_purchase_date_key	dim_date.date_key	Date the order was placed (no separate payment date exists in source).
customer_key	dim_customer.customer_key	Customer who paid.
payment_type_key	dim_payment_type.payment_type_key	Payment method.
order_status_key	dim_order_status.order_status_key	Current status of the parent order.

Measures

Measure	Type	Additivity	Definition
payment_value	NUMERIC(10,2)	Additive	Amount paid in this payment record (BRL). Not the same as item sales revenue.
payment_installments	SMALLINT	Non-additive	Number of instalments chosen (0 in 2 source rows, treated as 1).

Degenerate dimensions: order_id, payment_sequential.

4.4 fact_reviews (Transaction fact)

Business process	Customer reviews (satisfaction)
Grain	One row per review submitted for an order (review_id + order_id). The same review_id can cover more than one order, and an order can receive more than one review.
Primary key	review_id + order_id (degenerate source identifiers)
Source	olist_order_reviews_dataset.csv joined to olist_orders_dataset.csv for customer
Expected rows	99,224

Foreign keys

FK column	References	Meaning
customer_key	dim_customer.customer_key	Customer who wrote the review.
review_creation_date_key	dim_date.date_key	Date the satisfaction survey was sent (role: review date).
review_answer_date_key	dim_date.date_key	Date the customer answered (role: answer date).

Measures

Measure	Type	Additivity	Definition
review_score	SMALLINT	Non-additive	Satisfaction score 1 (worst) to 5 (best).
response_hours	NUMERIC(8,2)	Non-additive	Hours between survey creation and customer answer.
has_comment_title	SMALLINT	Additive	1 if the customer wrote a comment title.
has_comment_message	SMALLINT	Additive	1 if the customer wrote a comment message.

Degenerate dimensions: review_id, order_id.

5. Additivity rules

- **Additive** measures can be summed across every dimension. Examples are item_price, freight_value, payment_value, item_count and the 0/1 flags (is_late, is_delivered, has_comment_message). Flags are stored as SMALLINT so that SUM gives a count.
- **Non-additive** measures must be averaged or recomputed, never summed. Examples are review_score, avg_review_score, payment_installments, the day-duration measures (days_*), delivery_delay_days and customer_order_number. Ratios such as on-time rate and AOV are computed from additive parts.
- **Semi-additive** measures do not occur in this model, because there are no balance or stock snapshots.
- Order-level values (order_item_value, order_payment_value) are stored only in fact_orders, with one row per order. They must not be summed after joining to item or payment rows.

6. Dimension tables

6.1 dim_date

Type	Conformed, role-playing date dimension
Grain	One row per calendar day from 2016-01-01 to 2018-12-31, plus one 'Unknown / not occurred' row (date_key = -1).
Surrogate key	date_key (smart key YYYYMMDD)
Natural key	full_date
Attributes	day_of_month, day_of_week_num, day_of_week_name, is_weekend, week_of_year, month_num, month_name, year_month, quarter_num, year_quarter, year
Hierarchies	Year > Quarter > Month > Date; Year > Week of year > Date
Change handling	Static
Source	Generated in ETL (calendar); not a source file.
Used by	fact_order_items, fact_orders, fact_payments, fact_reviews (conformed)
Expected rows	1,096 + 1 unknown

6.2 dim_customer

Type	Conformed dimension (SCD Type 1)
Grain	One row per real customer (customer_unique_id), plus one 'Unknown' row (customer_key = -1).
Surrogate key	customer_key
Natural key	customer_unique_id
Attributes	customer_zip_code_prefix, customer_city, customer_state, customer_state_name, customer_region, customer_latitude, customer_longitude, first_purchase_date
Hierarchies	Region > State > City > Zip code prefix
Change handling	Type 1 - location taken from the customer's most recent order
Source	olist_customers_dataset.csv (+ olist_geolocation_dataset.csv for coordinates, olist_orders_dataset.csv for first purchase date)
Used by	fact_order_items, fact_orders, fact_payments, fact_reviews (conformed)
Expected rows	96,096 + 1 unknown

6.3 dim_product

Type	Dimension (SCD Type 1)
Grain	One row per product (product_id), plus one 'Unknown' row (product_key = -1).
Surrogate key	product_key
Natural key	product_id
Attributes	product_category_name_pt, product_category_name_en, product_name_length, product_description_length, product_photos_qty, product_weight_g, product_length_cm, product_height_cm, product_width_cm, product_volume_cm3, weight_band
Hierarchies	Product category > Product
Change handling	Type 1
Source	olist_products_dataset.csv + product_category_name_translation.csv
Used by	fact_order_items
Expected rows	32,951 + 1 unknown

6.4 dim_seller

Type	Dimension (SCD Type 1)
Grain	One row per seller (seller_id), plus one 'Unknown' row (seller_key = -1).
Surrogate key	seller_key
Natural key	seller_id
Attributes	seller_zip_code_prefix, seller_city, seller_state, seller_state_name, seller_region, seller_latitude, seller_longitude
Hierarchies	Region > State > City > Zip code prefix
Change handling	Type 1
Source	olist_sellers_dataset.csv (+ olist_geolocation_dataset.csv for coordinates)
Used by	fact_order_items
Expected rows	3,095 + 1 unknown

6.5 dim_payment_type

Type	Small descriptive dimension
Grain	One row per payment method code, plus one 'Unknown' row (payment_type_key = -1).
Surrogate key	payment_type_key
Natural key	payment_type_code
Attributes	payment_type_name, payment_method_group
Hierarchies	Payment method group > Payment type
Change handling	Type 1
Source	Distinct order_payments.payment_type + manual descriptions
Used by	fact_payments
Expected rows	5 + 1 unknown

6.6 dim_order_status

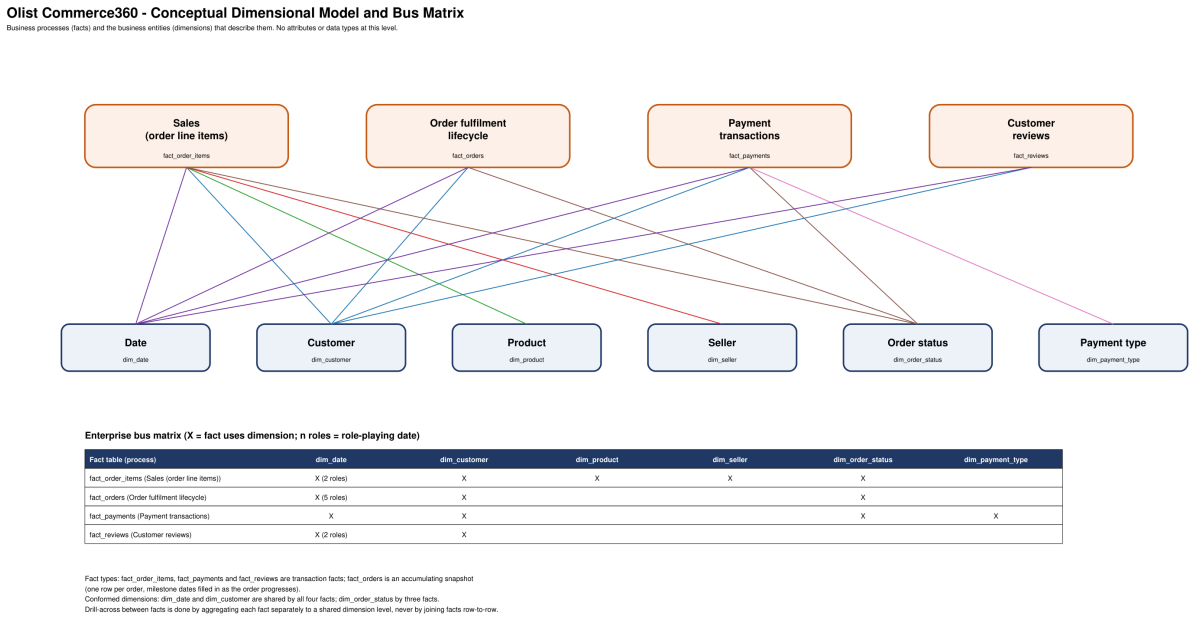
Type	Small descriptive dimension
Grain	One row per order status code, plus one 'Unknown' row (order_status_key = -1).
Surrogate key	order_status_key
Natural key	order_status_code
Attributes	order_status_name, status_group, is_final_status
Hierarchies	Status group > Order status
Change handling	Type 1
Source	Distinct orders.order_status + manual grouping
Used by	fact_order_items, fact_orders, fact_payments (conformed)
Expected rows	8 + 1 unknown

7. Conformed dimensions and bus matrix

dim_date and dim_customer are shared by all four facts, and dim_order_status is shared by three. Because the same dimension rows are used, results from different facts can be compared at a common level, such as month or customer state (drill-across). The facts themselves are never joined.

Fact	dim_date	dim_customer	dim_product	dim_seller	dim_payment_type	dim_order_status
fact_order_items	X (2 roles)	X	X	X		X
fact_orders	X (5 roles)	X				X
fact_payments	X	X			X	X
fact_reviews	X (2 roles)	X				

Figure 1 - Conceptual model (business processes, dimensions and bus matrix). Full size: P1_06_Star_Schema.pdf, page 2.



8. Date roles

Fact	Date FK (role)	Source column
fact_order_items	order_purchase_date_key	orders.order_purchase_timestamp
fact_order_items	shipping_limit_date_key	order_items.shipping_limit_date
fact_orders	order_purchase_date_key	orders.order_purchase_timestamp
fact_orders	approved_date_key	orders.order_approved_at
fact_orders	carrier_date_key	orders.order_delivered_carrier_date
fact_orders	delivered_date_key	orders.order_delivered_customer_date
fact_orders	estimated_delivery_date_key	orders.order_estimated_delivery_date
fact_payments	order_purchase_date_key	orders.order_purchase_timestamp
fact_reviews	review_creation_date_key	reviews.review_creation_date
fact_reviews	review_answer_date_key	reviews.review_answer_timestamp

dim_date is a single physical table used in several roles. In Power BI, one relationship per fact is active, normally the purchase date. The other roles are used through inactive relationships (USERELATIONSHIP) or role-specific views.

9. Natural keys, surrogate keys and unknown members

- Every dimension has an integer surrogate key generated in the warehouse (identity or sequence). dim_date uses a smart key in YYYYMMDD form. The source business key is kept as the natural key and enforced as UNIQUE.
- Fact rows are loaded by looking up each business key in its dimension and storing the surrogate key (for example customer_id -> customer_unique_id -> customer_key).
- Each dimension has an 'Unknown' row with key -1. Missing or unmatched keys, including dates that have not yet occurred, point to -1, so foreign keys are never NULL and inner joins do not lose fact rows.
- Fact tables use the source identifiers as degenerate primary keys: order_id, order_id + order_item_id, order_id + payment_sequential, and review_id + order_id. Profiling verified that each is unique.

10. Logical model

The logical model lists each entity with its keys and attributes, independent of PostgreSQL data types. The physical model, with data types, nullability and the full diagram, is in P1_06 and P1_07.

Table	Primary key	Foreign keys	Attributes / measures
dim_date	date_key	-	full_date, day_of_month, day_of_week_num, day_of_week_name, is_weekend, week_of_year, month_num, month_name, year_month, quarter_num, year_quarter, year
dim_customer	customer_key	-	customer_unique_id, customer_zip_code_prefix, customer_city, customer_state, customer_state_name, customer_region, customer_latitude, customer_longitude, first_purchase_date
dim_product	product_key	-	product_id, product_category_name_pt, product_category_name_en, product_name_length, product_description_length, product_photos_qty, product_weight_g, product_length_cm, product_height_cm, product_width_cm, product_volume_cm3, weight_band
dim_seller	seller_key	-	seller_id, seller_zip_code_prefix, seller_city, seller_state, seller_state_name, seller_region, seller_latitude, seller_longitude
dim_payment_type	payment_type_key	-	payment_type_code, payment_type_name, payment_method_group
dim_order_status	order_status_key	-	order_status_code, order_status_name, status_group, is_final_status
fact_order_items	order_id, order_item_id	order_purchase_date_key, shipping_limit_date_key, customer_key, product_key, seller_key, order_status_key	item_price, freight_value, item_gross_value, item_quantity
fact_orders	order_id	customer_key, order_status_key, order_purchase_date_key, approved_date_key, carrier_date_key, delivered_date_key, estimated_delivery_date_key	purchase_hour, item_count, order_item_value, order_freight_value, order_payment_value, payment_count, days_purchase_to_approval, days_approval_to_carrier, days_carrier_to_customer, days_purchase_to_delivery, days_promised, delivery_delay_days, is_delivered, is_late, customer_order_number, is_repeat_purchase, review_count, avg_review_score
fact_payments	order_id, payment_sequential	order_purchase_date_key, customer_key, payment_type_key, order_status_key	payment_value, payment_installments
fact_reviews	review_id, order_id	customer_key, review_creation_date_key, review_answer_date_key	review_score, response_hours, has_comment_title, has_comment_message

11. Schema choice: star schema

A **star schema** is used, with four fact tables sharing conformed dimensions (a fact constellation). Each dimension joins directly to the facts, with no dimension-to-dimension joins. The reasons are:

- The hierarchies are short and fixed: Region > State > City > Zip prefix, Category > Product, and Year > Quarter > Month > Date. Normalising them would add joins without a meaningful storage saving.
- The category translation is a 1:1 lookup with 71 rows. Folding it into dim_product removes a snowflake table.
- Power BI performs best and is easiest to understand with a star schema of single-direction 1:* relationships.
- OLAP queries (ROLLUP/CUBE over region, state and category) need only one join per dimension.
- The trade-off is that region and state values repeat in dim_customer (about 96,000 rows), which is negligible at this data volume.